

Assignment 1

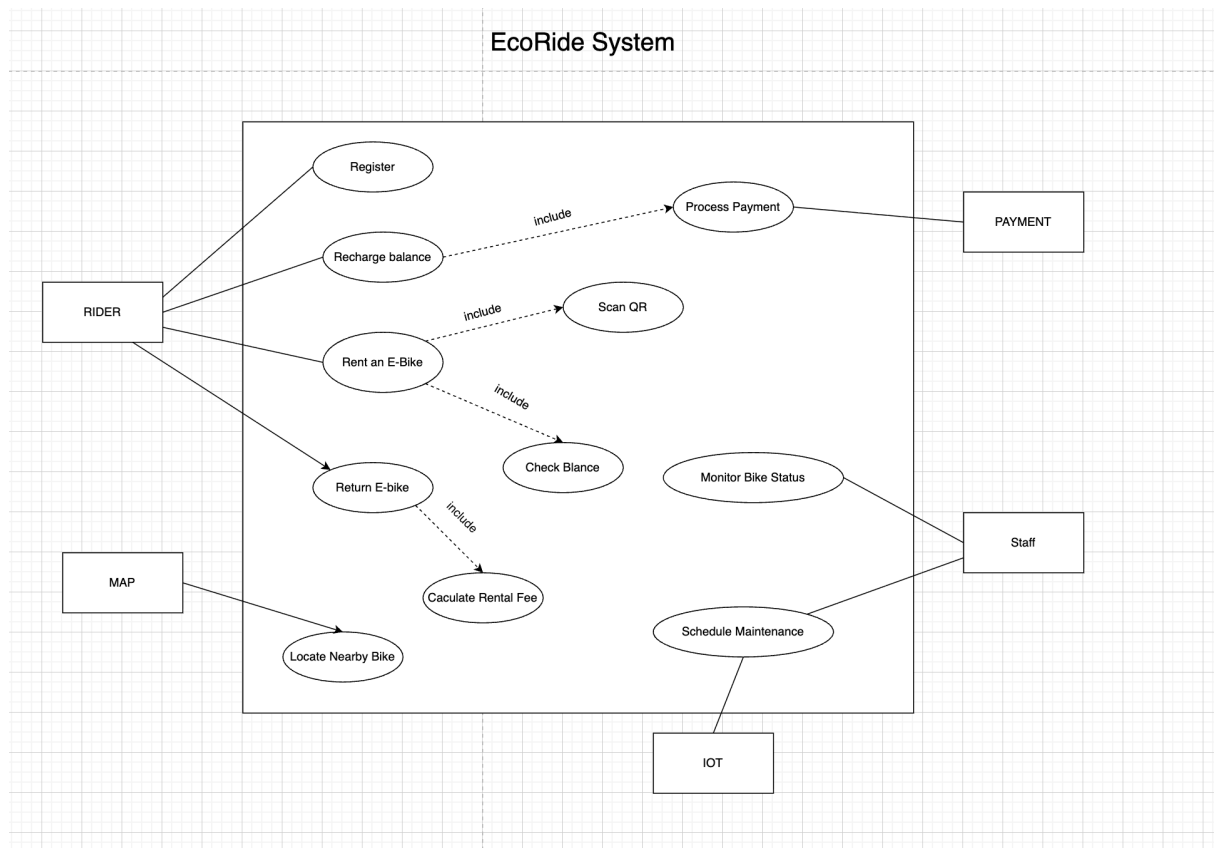
Question 1:

1.1: Actor: Renter, Staff, Payment, Map, Iot

1.2: UC:

- + register account
- + recharge balance
- + rent E-bike
- + calculate rent fee
- + monitor status
- + scheduler maintenance

1.3: Diagram



1.4:

Actor	Role
User	Rent and use e-bikes
Staff	Manage fleet and maintenance

Payment	Process financial transactions
lot	Provide bike locations
map	Send bike telemetry data

Question 2:

2.1

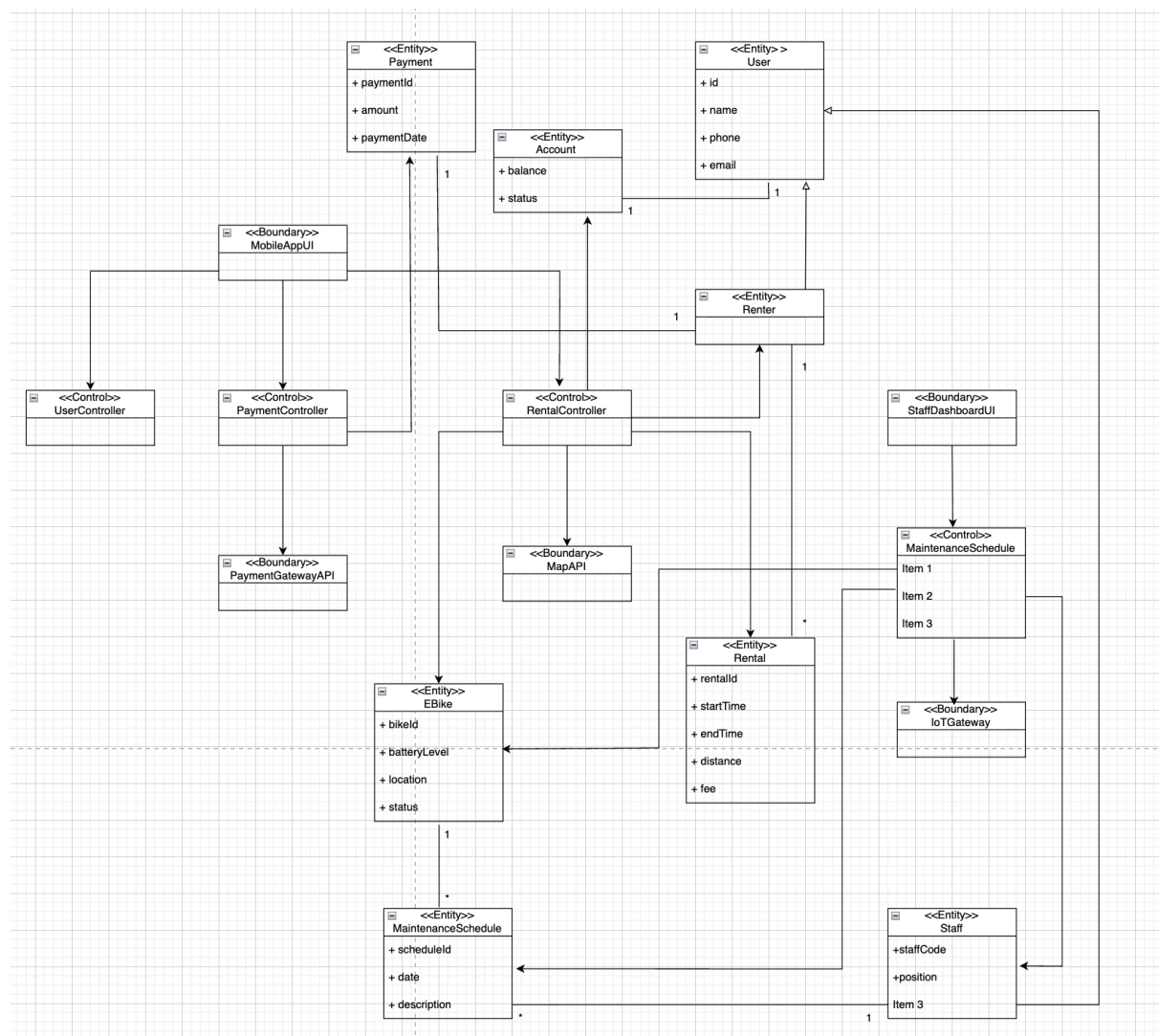
Entity Classes: User, Account,EBike,Rental,Payment,MaintenanceSchedule

Boundary Classes: MobileAppUI, PaymentGatewayAPI, MapAPI, IoTGateway

Control Classes: UserController, RentalController, PaymentController,

MaintenanceController

2.2



2.3

Entity: Business Data Storage

Boundary: External Communication

Control: Business Flow Coordination

Question 4:

4.1 Architectural Decision: Why SOA is the Ideal Choice for EcoRide. The Selection of Service-Oriented Architecture (SOA) for the EcoRide system is driven by two critical architectural requirements: seamless third-party integration and modular scalability.

4.2 Major System:

1. Mobile App Service

- User Interface
- Ride Request Submission

2. User Management Service

- Registration
- Authentication / Login
- Account Management

3. Rental Service

- Vehicle Rental
- Vehicle Return
- Fare Calculation

4. Payment Service

- Top-up
- Payment Processing

5. Bike Monitoring Service

- Battery Tracking
- Location Tracking

6. Maintenance Service

- Maintenance Scheduling

7. External Services

- Payment Gateway
- Map Service
- IoT Tracking Platform